

Matplotlib Notes(DAA/JDA)

1. Introduction to Matplotlib

Matplotlib is a popular Python library used for creating graphs, charts, and visual representations of data. It helps users understand and analyze data easily through graphical visualization.

Features

- Easy to use
- Open Source
- Supports multiple chart types
- Highly customizable
- Works with NumPy and Pandas

Import Matplotlib

```
import matplotlib.pyplot as plt
```

2. Line Plot

A Line Plot displays data points connected by straight lines. It is used to show trends, comparisons, and changes over time.

Syntax

```
plt.plot(x, y)
```

Example

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4]
y = [10, 20, 15, 25]
```

```
plt.plot(x, y)
plt.show()
```

Common Parameters

Parameter	Description
marker	Symbol at data points

color (c)	Line color
linestyle (ls)	Line style
linewidth (lw)	Line thickness

3. Markers

Markers are symbols used to highlight data points on a graph.

Example

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4]
y = [10, 20, 15, 25]
```

```
plt.plot(x, y, marker='o')
plt.show()
```

Common Marker Types

Marker	Meaning
o	Circle
*	Star
.	Point
+	Plus
x	Cross
s	Square
^	Triangle

4. Line Styles

Line styles change the appearance of the line.

Example

```
plt.plot(x, y, linestyle='--')
plt.show()
```

Styles

Style	Meaning
-	Solid
--	Dashed
-.	Dash Dot
:	Dotted

5. Colors

Colors make graphs attractive and easy to understand.

Example

```
plt.plot(x, y, color='red')
plt.show()
```

Shortcut Format

```
plt.plot(x, y, 'o--r')
```

Meaning:

- o → Circle Marker
- -- → Dashed Line
- r → Red Color

Common Colors

- red
 - blue
 - green
 - yellow
 - orange
 - purple
 - black
 - pink
-

6. Marker Customization

Markers can be customized using size, face color, and edge color.

Example

```
plt.plot(
    x, y,
    marker='*',
    ms=12,
    mfc='red',
    mec='green'
)
plt.show()
```

Parameters

Parameter	Description
ms	Marker Size
mfc	Marker Face Color
mec	Marker Edge Color

7. Labels and Title

Labels identify the axes and the title describes the graph.

Example

```
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Simple Line Graph")

plt.plot(x, y)
plt.show()
```

Title Position

```
plt.title("Simple Line Graph", loc="center")
```

Possible values:

- left
 - center
 - right
-

8. Font Customization

Font dictionaries are used to customize labels and titles.

Example

```
f1 = {  
    "family": "Arial",  
    "color": "green",  
    "size": 20  
}
```

```
plt.title("Simple Line Graph", fontdict=f1)
```

Font Properties

Property	Description
family	Font Family
size	Font Size
color	Font Color

9. Grid

Grid lines improve readability by making values easier to compare.

Example

```
plt.plot(x, y)  
plt.grid()  
plt.show()
```

Customized Grid

```
plt.grid(  
    color="green",  
    linestyle="--",  
    linewidth=0.5  
)
```

Grid Parameters

Parameter	Description
color	Grid Color

linestyle	Grid Style
linewidth	Thickness
axis	x, y or both

10. Subplots

Subplots allow multiple graphs to be displayed in one figure.

Syntax

```
plt.subplot(rows, columns, position)
```

Example

```
import matplotlib.pyplot as plt
```

```
plt.subplot(2,2,1)  
plt.plot([1,2,3],[10,20,30])
```

```
plt.subplot(2,2,2)  
plt.plot([1,2,3],[30,20,10])
```

```
plt.show()
```

Benefits

- Compare multiple datasets
 - Save display space
 - Better analysis
-

11. Scatter Plot

A Scatter Plot displays data using individual dots. It helps identify relationships between two variables.

Syntax

```
plt.scatter(x, y)
```

Example

```
import matplotlib.pyplot as plt
```

```
age = [18,20,22,24]
```

```
salary = [15000,18000,22000,26000]
```

```
plt.scatter(age, salary)
```

```
plt.xlabel("Age")  
plt.ylabel("Salary")  
plt.title("Age vs Salary")
```

```
plt.show()
```

Applications

- Age vs Salary
 - Height vs Weight
 - Sales vs Profit
 - Student Performance Analysis
-

12. Multiple Scatter Plots

Multiple scatter plots are used to compare datasets.

Example

```
age_a = [22,24,26,28]  
salary_a = [18000,22000,26000,30000]
```

```
age_b = [22,24,26,28]  
salary_b = [16000,20000,24000,28000]
```

```
plt.scatter(age_a, salary_a,  
            label="Company A",  
            color="red")
```

```
plt.scatter(age_b, salary_b,  
            label="Company B",  
            color="green")
```

```
plt.legend()  
plt.show()
```

13. Legend

Legend identifies different datasets in a graph.

Example

```
plt.legend()
```

Output

- Company A
 - Company B
-

14. Colors in Scatter Plot

Colors can be assigned manually to each point.

Example

```
colors = ["red","green","blue","orange"]
```

```
plt.scatter(age, salary,  
            color=colors)
```

```
plt.show()
```

15. Colormap (cmap)

A Colormap automatically assigns colors based on data values.

Example

```
experience = [1,2,3,4,5,6]
```

```
plt.scatter(  
    age,  
    salary,  
    c=experience,  
    cmap="rainbow"  
)
```

```
plt.show()
```

Popular Colormaps

Colormap
rainbow
Reds
Blues

Greens
viridis
plasma
jet
coolwarm
spring
summer
winter
autumn

Benefits

- Better visualization
 - Represents third variable
 - Easy comparison of values
-

16. Colorbar

A Colorbar explains the meaning of colors in a graph.

Example

```
sc = plt.scatter(  
    age,  
    salary,  
    c=experience,  
    cmap="Reds"  
)  
  
plt.colorbar(  
    sc,  
    label="Experience"  
)  
  
plt.show()
```

Purpose

- Shows color scale
 - Helps understand color values
-

17. Marker Size (s)

The `s` parameter controls the size of scatter plot points.

Example

```
sizes = [50,100,150,200,250,300]
```

```
plt.scatter(  
    age,  
    salary,  
    s=sizes  
)
```

```
plt.show()
```

Uses

- Highlight important data
 - Represent another variable visually
-

18. Alpha (Transparency)

Alpha controls the transparency of points or objects.

Example

```
plt.scatter(  
    age,  
    salary,  
    alpha=0.5  
)
```

```
plt.show()
```

Alpha Values

Value	Result
0	Fully Transparent

0.5	Semi Transparent
1	Fully Visible

Advantages

- Avoids overlapping issues
 - Improves visualization
-

19. Complete Scatter Plot Example

```
import matplotlib.pyplot as plt
```

```
age = [18,20,22,24,26,28]
```

```
salary = [15000,18000,22000,26000,30000,35000]
```

```
experience = [1,2,3,4,5,6]
```

```
sizes = [50,100,150,200,250,300]
```

```
sc = plt.scatter(  
    age,  
    salary,  
    c=experience,  
    cmap="jet",  
    s=sizes,  
    alpha=0.8  
)
```

```
plt.xlabel("Age")
```

```
plt.ylabel("Salary")
```

```
plt.title("Age vs Salary with Experience")
```

```
plt.colorbar(sc,  
    label="Experience")
```

```
plt.grid()
```

```
plt.show()
```